
SOFTWARE ENGINEERING

LAB 3

Component Modelling and Architectural Pattern Selection

Digital Campus Library Reservation Gateway

Problem Statement #08
Campus & Academic Operations

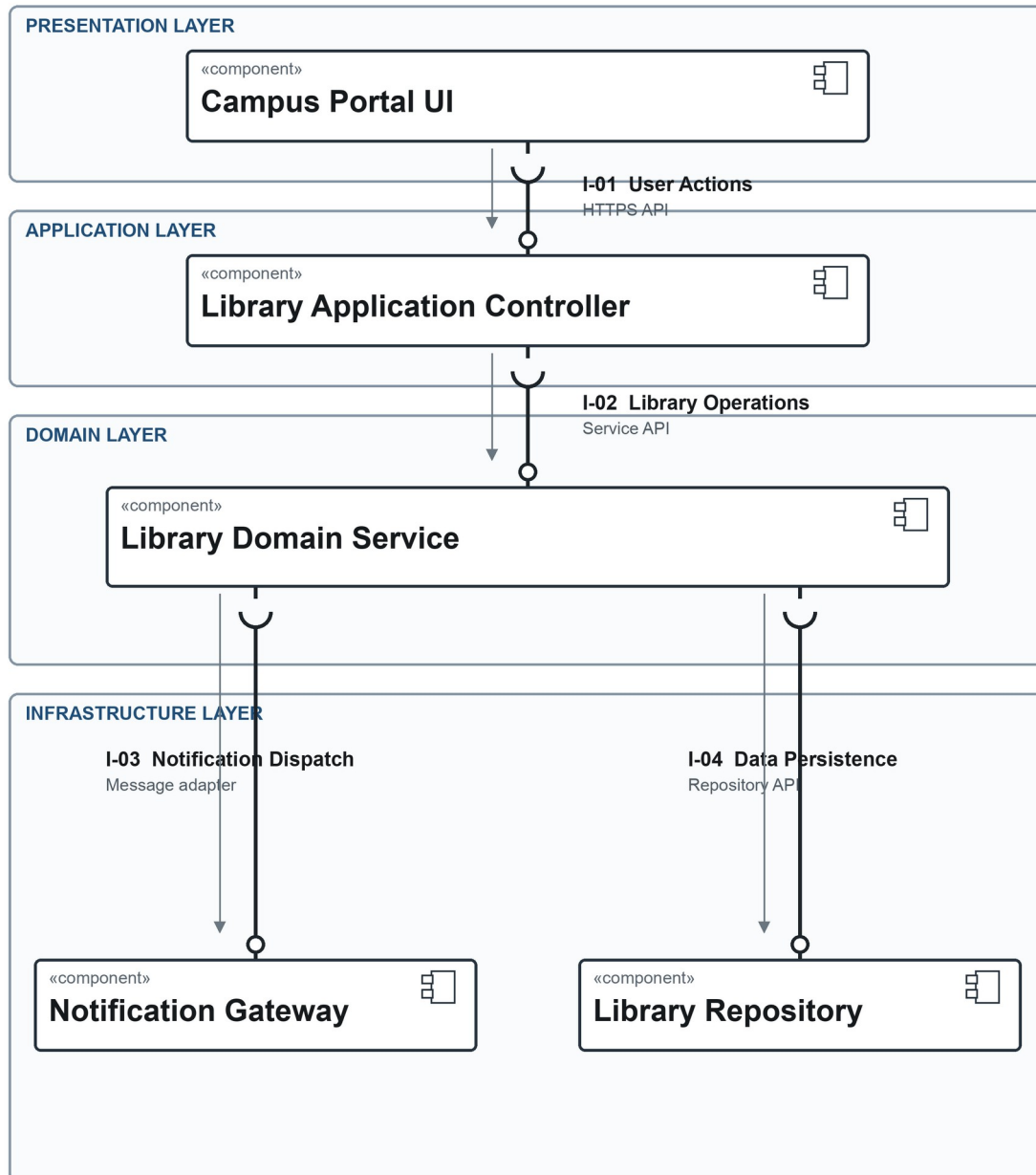
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UML Component Diagram

Selected pattern: Layered Architecture. The diagram contains exactly five components and four internal interfaces, using provided-interface (ball) and required-interface (socket) notation.

Digital Campus Library Reservation Gateway

Layered Architecture — UML Component Diagram



Exactly 5 components • Exactly 4 interfaces • Ball = provided • Socket = required

Figure 1. Component model for the Digital Campus Library Reservation Gateway.

Architectural Justification

We chose Layered Architecture for the Digital Campus Library Reservation Gateway.

Architecture Selection

The gateway is divided into presentation, application, domain and infrastructure layers. The Campus Portal UI handles Student Member and Head Librarian interactions. The Library Application Controller authenticates requests and coordinates use cases. The Library Domain Service owns catalogue, circulation, FIFO reservation and fine rules. The Notification Gateway and Library Repository isolate message delivery and persistent storage.

Reason 1 — Reliable FIFO reservations

Placing a hold, returning a copy, promoting the queue head and assigning a collection deadline affect shared state. Keeping these rules in one domain layer allows the operation to be completed transactionally. Concurrent requests cannot receive the same position, bypass an earlier member or offer a returned copy to the wrong person.

Reason 2 — Appropriate for the system's scale

The system serves one campus catalogue of approximately 50,000 titles and follows a direct workflow: authenticate, execute a library rule, persist the result and respond. In-process layers are easier to deploy, test and maintain than separate microservices. A Client–Server arrangement can still be used externally, but it does not adequately organize the server's internal responsibilities.

Security Advantage

The Application Controller acts as the trust boundary. All requests use authenticated, encrypted sessions, and role checks separate Student Member operations from Head Librarian catalogue changes. The UI cannot directly modify holds, loans, fines or ISBN records, reducing unauthorized changes and preventing storage details from being exposed.

Performance Benefit

Indexed ISBN, title and author fields, together with pagination, avoid scanning or returning the full catalogue. Internal components communicate without network hops, helping searches across 50,000 titles remain below the 150 ms target. Notification delivery is separated from the return transaction so that a slow messaging channel does not delay circulation processing.

Reliability Note

Confirmed holds, sequence numbers and pending notification events are written to durable storage before success is returned. After a restart, the same queue order can be reconstructed and unsent notifications retried safely, supporting the 99.5% availability requirement.

Conclusion

Layered Architecture provides the clearest balance of fairness, speed, security and maintainability for the requirements prepared in Lab 1 and planned in Lab 2.